

Results

Experimental scope

Item	Value
Workload	TinyLlama-1.1B
Transformer layers	22
GPUs	8
Pipeline stages	2
TP/DP strategy pairs	9
Layer partitions	21
Evaluated configurations	189

The retained artifacts identify the GPU count but not the GPU model, so no accelerator model or memory capacity is claimed here.

Estimator error

Mean absolute prediction error across all 189 recorded configurations was **5.85%**, using measured net time as the denominator.

Stage 0	Stage 1	Mean	Minimum	Maximum
TP2_DP2	TP2_DP2	5.13%	2.71% (17/5)	8.74% (9/13)
TP4_DP1	TP4_DP1	5.37%	2.63% (19/3)	9.99% (11/11)
TP4_DP1	TP2_DP2	5.42%	2.62% (21/1)	9.74% (9/13)
TP4_DP1	TP1_DP4	5.52%	2.65% (20/2)	10.88% (9/13)
TP2_DP2	TP1_DP4	5.75%	2.77% (20/2)	9.89% (7/15)
TP2_DP2	TP4_DP1	5.78%	2.84% (21/1)	9.54% (10/12)
TP1_DP4	TP1_DP4	6.42%	3.92% (19/3)	9.30% (7/15)
TP1_DP4	TP4_DP1	6.48%	4.50% (1/21)	9.88% (11/11)
TP1_DP4	TP2_DP2	6.75%	3.59% (19/3)	10.87% (11/11)

Selected plan

The minimum estimated latency was obtained with an even 11/11 layer split and **TP1_DP4** in both stages:

Stage	Layers	Strategy	Estimated stage cost
0	[0, 11)	TP1_DP4	593.25 ms
1	[11, 22)	TP1_DP4	594.60 ms

The estimated 1F1B latency was 5,350.04 ms for eight microbatches. This was a **homogeneous** plan. TinyLlama's relatively uniform Transformer blocks did not make a heterogeneous plan optimal in this experiment.

The supported conclusion is therefore not that heterogeneity improved performance. The contribution was expanding the runtime and search space so heterogeneous plans could be executed, modeled, and compared.

Measurement limitations

- The experiment covered one model and one eight-GPU scale point.
- Accelerator model and interconnect metadata were not retained in a publication-safe artifact.
- The archived prototype used a leader-based activation replication path; the experiment did not establish unique-sample data-parallel scaling.
- Raw rank logs are excluded from the public reconstruction pending ownership and privacy review.
- The estimator assumes repeated blocks have stable marginal cost.